

US Air Force confirms first successful AI dogfight

theverge.com/2024/4/18/24133870/us-air-force-ai-dogfight-test-x-62a

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An autonomously controlled aircraft faced off against a human pilot in a test last year.

By Emma Roth, a news writer who covers the streaming wars, consumer tech, crypto, social media, and much more. Previously, she was a writer and editor at MUO.

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Image: DARPA

The US Air Force is putting AI in the pilot's seat. In an update on Thursday, the Defense Advanced Research Projects Agency (DARPA) revealed that an AI-controlled jet successfully faced a human pilot during an in-air dogfight test carried out last year.

DARPA began experimenting with AI applications in December 2022 as part of its Air Combat Evolution (ACE) program. It worked to develop an AI system capable of autonomously flying a fighter jet, while also adhering to the Air Force's safety protocols.



Watch Video At: <https://youtu.be/C8yjtaZfNCw>

After carrying out dogfighting simulations using the AI pilot, DARPA put its work to the test by installing the AI system inside its experimental X-62A aircraft. That allowed it to get the AI-controlled craft into the air at the Edwards Air Force Base in California, where it says it carried out its first successful dogfight test against a human in September 2023.

Human pilots were on board the X-62A with controls to disable the AI system, but DARPA says the pilots didn't need to use the safety switch "at any point." The X-62A went against an F-16 controlled solely by a human pilot, where both aircraft demonstrated "high-aspect nose-to-nose engagements" and got as close as 2,000 feet at 1,200 miles per hour. DARPA doesn't say which aircraft won the dogfight, however.

"Dogfighting was the problem to solve so we could start testing autonomous artificial intelligence systems in the air," Bill Gray, the chief test pilot at the Air Force's Test Pilot School, said in a statement. "Every lesson we're learning applies to every task you could give to an autonomous system."

The agency has conducted a total of 21 test flights so far and says the tests will continue through 2024. Rapid advancements in AI have given rise to concerns over how the military might use the systems. *The Wall Street Journal* reported last year that the Pentagon is looking to develop AI systems for defense and to enhance its fleet of drones.